

B.2 Greenhouse effect

Topic warm-up · IB Physics (2023 syllabus)

with worked answers

Name: _____

Date: _____

Total: **24 marks**

Show your working.

Question 1 — Planet Flat White

(6 marks)

On a cloudless noon, radiation of intensity 1000 W m^{-2} arrives at the top of the atmosphere directly above a coastal town. The local atmosphere and surface reflect with a combined albedo of 0.35.

- (a) Define albedo. [1]
- (b) Calculate the intensity absorbed by the town below. [2]
- (c) A student answers 350 W m^{-2} . State precisely which quantity the student has calculated instead. [1]
- (d) Give two reasons why the Earth's average albedo varies from place to place and season to season. [2]

Question 2 — The quartered Sun

(5 marks)

The solar constant (intensity arriving at the top of the atmosphere, facing the Sun) is $S = 1360 \text{ W m}^{-2}$, yet climate diagrams label the *average* incoming radiation as $S/4 = 340 \text{ W m}^{-2}$.

- (a) Explain the factor of 4, using the areas πR^2 and $4\pi R^2$. [3]
- (b) State when it is an error to apply this factor of 4 to an intensity problem. [2]

Question 3 — The gas line-up

(8 marks)

Four atmospheric gases stand accused of warming the planet: methane, water vapour, nitrogen, carbon dioxide.

- (a) Identify the innocent gas, despite it making up most of the atmosphere. [1]
- (b) Explain what a molecule needs in order to act as a greenhouse gas, and hence why your answer to (a) does not qualify. [3]
- (c) Describe the greenhouse mechanism as an energy story: what the atmosphere lets in, what the surface emits, and what the greenhouse gases do with it. [4]

Question 4 — Natural versus enhanced

(5 marks)

- (a) Distinguish between the natural greenhouse effect and the *enhanced* greenhouse effect. [2]
- (b) Explain the ice–albedo feedback: what melting sea ice does to the energy balance that melted it. [3]

Answers

Question 1 — Planet Flat White

- (a) The fraction of the incident radiation that is reflected, or scattered back, rather than absorbed.
- (b) $(1 - 0.35) \times 1000 = 650 \text{ W m}^{-2}$.
- (c) The *reflected* intensity, albedo $\times$ incident (0.35×1000), rather than the absorbed intensity.
- (d) Any two of: snow and ice cover; cloud cover; vegetation against desert against ocean; the season, through the Sun's angle.

Question 2 — The quartered Sun

- (a) The Earth intercepts sunlight across its disc, of area πR^2 , but spreads that energy over the whole rotating sphere, area $4\pi R^2$, taking day, night and all latitudes together. The ratio of the two areas is 4, so the average incoming intensity is $S/4$.
- (b) Whenever the question is *local and instantaneous*, for example the Sun directly overhead at noon. Then the full S applies (times $1 - \text{albedo}$), not $S/4$.

Question 3 — The gas line-up

- (a) Nitrogen.
- (b) It must be able to absorb infrared, which requires vibrational modes that change the molecule's dipole moment. N_2 is a symmetric diatomic molecule with no such modes, so it is effectively transparent to infrared.
- (c) The atmosphere is largely transparent to incoming short-wave (visible) radiation, which reaches and warms the surface. The warmed surface re-emits at much longer infrared wavelengths. Greenhouse gases absorb this infrared and re-radiate it in all directions, so a share returns to the surface. The surface therefore settles at a higher equilibrium temperature than it would with a transparent atmosphere.

Question 4 — Natural versus enhanced

- (a) Natural: the pre-industrial effect, roughly 33 K of warming, produced by naturally occurring greenhouse gases. Enhanced: the *additional* warming from gases added by human activity (CO_2 , CH_4 , N_2O).
- (b) Ice is highly reflective, so it has a high albedo. Melting it exposes dark ocean beneath, lowering the albedo, so more radiation is absorbed. That causes further warming and further melting: a positive feedback loop, in which the effect reinforces its own cause.